

SAI KIRAN BANDI

Data Engineer

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PROFILE SUMMARY

Highly accomplished and results-driven **Data Engineer** with **6+ years** of extensive experience in designing, implementing, and optimizing robust **data solutions** and scalable **ETL/ELT pipelines**. Proven expertise across **AWS (EC2, S3, RDS, Lambda, EMR, Athena, Glue)**, **Azure (SQL DW, Data Lake, Data Factory, Data Bricks with PySpark)**, and **Google Cloud Platform (GCP)**, including **Snowflake** cloud data warehousing. Skilled in **Apache Spark (PySpark, Spark SQL, Streaming)**, **Kafka**, **Hadoop distributions**, and workflow orchestration with **Apache Airflow** and Control-M. Adept at **CI/CD automation** using Jenkins,Docker,Git and Kubernetes, consistently delivering data-driven insights in **Agile Scrum** environments.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (S3, EC2, EMR, Redshift, RDS, Lambda, Kinesis, SNS, SQS, IAM, CloudFormation), Azure (Data Factory, Data Lake Storage, Databricks, SQL Database, SQL Data Warehouse, Analysis Services, HDInsight), GCP (Dataproc, BigQuery, Cloud Dataflow, Cloud Storage, Pub/Sub, Cloud Shell)
- **Big Data Technologies:** HDFS, Hue, MapReduce, Hive, HBase, Sqoop, Impala, Flume, Kafka, Spark (PySpark, Spark SQL, Spark Streaming, Spark Scala APIs, Spark MLlib, GraphX), Yarn, Cloudera Manager, Kerberos
- **Databases:** Oracle, Microsoft SQL Server, MySQL, DB2, Teradata, MongoDB, Cassandra, Snowflake
- **Programming Languages:** Python, Java, Scala, SQL, T-SQL, C++, JavaScript, jQuery
- **ETL/BI Tools:** Talend, AWS Glue, SSIS, SSRS, Informatica, Data Stage, Power BI, Tableau, AWS QuickSight, Data Studio, Elasticsearch, Kibana
- **DevOps & Automation:** Jenkins, Git, Terraform, Ansible, Docker, Kubernetes, Control-M, Oozie, Puppet, UNIX Shell Scripting
- **Methodologies:** Agile (Scrum), Waterfall, RUP, SDLC, Dimensional Modeling, Data Migration, Data Cleansing, Data Profiling
- **Web Technologies:** Django, Bootstrap, Angular, XML, HTML5, CSS3, JSON, RESTful Web Services, SOAP Web Services
- **IDEs:** Eclipse, Dreamweaver, PyCharm, Sublime Text
- **Operating Systems:** Windows ,Linux,Unix

EXPERIENCE

CHEMOURS, Wilmington, Delaware, USA

APR 2024 - Present

Azure Data Engineer

- Developed modern data solutions using **Azure PaaS** services to enable advanced data visualization and business insights, which increased efficiency in decision-making by 30%.
- Optimized workflows and minimized downtime by analyzing production environments to assess the impact of new installations on existing business processes.
- Implemented **Azure Data Factory (ADF)** to ingest data from diverse source systems, improving data accessibility and reducing processing time by 30%.
- Performed complex data transformations using Spark, storing results in **HDFS and Snowflake** for further analysis, which improved data accuracy and reporting capabilities
- Implemented **Kafka** with **Spark** streaming to create real-time solutions, significantly boosting data processing speed and enabling quicker decision-making.
- Enhanced data processing efficiency and accelerated insights by implementing big data integration and analytics with **Hadoop, SOLR, Spark, Kafka, Storm, and webMethods**.
- Developed efficient ETL pipelines in **Azure Databricks** with notebooks and Python, improving data processing speeds by 30% and reducing latency using Spark **DataFrames** and Spark SQL.
- Developed dynamic **Tableau** dashboards by integrating multiple data sources, significantly improving data visibility and empowering strategic decision-making efficiency.
- Developed data pipelines with Spark Scala APIs, **Hive-based data aggregation**, and **JSON** formatting for visualization, which improved data accuracy and reporting capabilities
- Designed and implemented **Infrastructure as Code (IaC)** using **Terraform** for automated Azure resource provisioning, reducing deployment time and minimizing manual errors
- Participated in all phases of the **Software Development Lifecycle (SDLC)**, including requirements gathering, design, development, deployment, and analysis.
- Managed large datasets using **Pandas** and **SQL** for efficient data processing and analysis.
- Ensured data integrity during migration by resolving compatibility issues with **T-SQL** scripting.

- Integrated **Azure Synapse** with Databricks, reducing development effort by 50% and improving loading performance through dynamic partitioning.
- Implemented end-to-end **CI/CD pipelines** using **Jenkins** for seamless deployment and release automation.

BNY Mellon Trust of Delaware, Wilmington, Delaware, USA

May 2023 – Mar 2024

AWS Data Engineer

- Developed and maintained scalable ETL pipelines using **AWS Glue**, **Lambda**, and **Step Functions** to automate data ingestion and transformation from diverse sources into **Amazon Redshift**, improving data processing efficiency by 35%.
- Built and optimized real-time data pipelines using **Apache Kafka** and **Spark Streaming** to enable faster insights and support critical financial reporting systems.
- Implemented data lake solutions on **Amazon S3** with efficient partitioning and **Parquet file formats**, reducing storage costs by 20% and enhancing query performance.
- Created **Redshift Spectrum** queries for seamless access to both structured and semi-structured data, improving data accessibility for downstream analytics and reporting.
- Designed and deployed **infrastructure-as-code (IAC)** using **AWS CloudFormation** and Terraform to provision secure and scalable cloud environments.
- Developed and maintained **AWS Glue jobs** and **crawlers** to **catalog** metadata and ensure accurate schema detection for downstream processing.
- Integrated **Azure Monitor** and **Log Analytics** to track pipeline performance, troubleshoot failures, and ensure high availability of data processing workflows.
- Automated CI/CD pipelines using **AWS CodePipeline** and **CodeBuild**, reducing deployment time by 30% and ensuring consistent environment provisioning.
- Enhanced data quality through **AWS Data Quality** rules and custom **Python scripts**, ensuring 99.9% data accuracy and integrity across pipelines.
- Collaborated with cross-functional teams to gather business requirements, design data models, and implement analytics solutions using **Redshift**, **QuickSight**, and **Tableau**.
- Monitored and troubleshooting pipeline performance using **AWS CloudWatch** and **X-Ray**, reducing data processing failures by 25% through proactive alerts and diagnostics.

Murugappa Group, Chennai, INDIA

Jun 2021- Dec 2022

GCP Data Engineer

- Designed and implemented scalable ETL pipelines using **Google Cloud Dataflow**, **Cloud Composer (Airflow)**, and **Big-Query** to streamline data ingestion, transformation, and analytics workflows, reducing processing time by 40%.
- Developed batch and real-time data pipelines using **Apache Beam** and **PySpark** on **Google Cloud DataProc**, improving data processing efficiency by 30%.
- Built and managed secure, cost-effective data lake solutions on **Google Cloud Storage (GCS)** with optimized partitioning and **Parquet file formats**, reducing storage costs by 25%.
- Created and optimized **Big-Query** datasets, views, and materialized tables for advanced analytics and reporting, enhancing query performance by 50%.
- Implemented CI/CD pipelines using **Cloud Build** and GitHub Actions, reducing deployment time and ensuring consistent environments across development, testing, and production.
- Developed and scheduled **Cloud Composer DAGs** for data pipeline orchestration, ensuring timely execution and monitoring of end-to-end workflows.
- Automated data quality checks using **Google Data Quality** and custom **Python scripts**, ensuring 99.9% data accuracy and integrity.
- Built interactive dashboards using **Google Data Studio** and **Looker**, enabling real-time insights for **business stakeholders** and driving data-driven decision-making.
- Monitored pipeline performance using **Google Cloud Operations Suite (Stack-driver)**, proactively identifying and resolving **bottlenecks**, reducing data pipeline failures by 20%.
- Collaborated with **cross-functional teams** to gather business requirements, design data models, and implement end-to-end data solutions supporting enterprise analytics and reporting.

Data Engineer

- Developed and managed relational and non-relational databases using **MySQL**, **PostgreSQL**, and **MongoDB**, ensuring optimal performance, data integrity, and security across production environments.
- Developed and optimized complex **SQL queries**, **stored procedures**, and **triggers for data extraction, transformation, and loading (ETL)**, improving query performance by 40%.
- Designed efficient indexing strategies and performed database partitioning, resulting in a 30% improvement in query execution speed for large datasets.
- Implemented database backup and recovery strategies using native tools and custom scripts, ensuring business continuity and minimizing data loss risks.
- Conducted database performance tuning through query optimization, indexing, and execution plan analysis, reducing database latency and improving system efficiency.
- Built real-time data replication pipelines using **AWS Database Migration Service (DMS)** and **Change Data Capture (CDC)** for cross-database synchronization.
- Developed and maintained **data marts** and **OLAP cubes** in **SQL Server Analysis Services (SSAS)** to support advanced analytics and business intelligence reporting.
- Collaborated with business analysts and data scientists to design normalized and **denormalized data models**, supporting efficient querying and reporting workflows.
- Monitored database health, identified performance **bottlenecks**, and implemented proactive solutions using database monitoring tools such as **AWS CloudWatch** and **Azure Monitor**

PROJECTS**1. Enterprise Sales Insights Platform (Talend + Snowflake + Tableau)**

- Built a secure, scalable **ETL framework** using **Talend** to ingest and transform sales data from ERP, CRM, and third-party APIs.
- Loaded curated data into **Snowflake** with **partitioning strategies to optimize dashboard performance demonstrating experience with transforming data from flat to normalized databases**.
- Integrated with **Tableau** for real-time executive reporting and trend forecasting.
- Reduced manual reporting by 75% and improved decision-making agility.

2. Real-Time IoT Data Pipeline (Kafka + Spark + AWS Glue)

- Designed and deployed a real-time data ingestion pipeline using **Kafka** to capture sensor data from manufacturing units.
- Processed and enriched streams with **Spark Streaming**, then persisted them into **AWS S3** in Parquet format via **AWS Glue**.
- Ensured secure data governance through **IAM roles** and **S3 bucket policies**.
- Enabled anomaly detection and downtime prevention via real-time monitoring dashboards.

EDUCATION**Master of Science in Information Systems Technologies, Wilmington University**

Delaware, USA

- **Relevant Course Work:** Data Management, Cloud Management, IT policy and strategy, Web and Data Security, Analysis, Modeling, Design

CERTIFICATIONS

- **AWS Certified Data Engineer - Associate**
- **Google Cloud Certified - Associate Cloud Engineer**